

Problem Statement

Localized Agronomic Knowledge Retrieval in Nigeria

Smallholder farmers in Nigeria produce much of the nation's food supply, yet they often struggle to get timely, accurate agronomic advice when they need it most. Crop diseases, pest infestations, soil degradation, and unpredictable climate conditions frequently go unaddressed. This is not because the knowledge does not exist, but because it is scattered across extension documents, research publications, and government guidelines that farmers have no easy way to search. A farmer describing a discolored leaf, a wilting stem, or dry soil in their own words often cannot find the right guidance, since keyword search fails to bridge the gap between how farmers describe problems and how experts document solutions. Limited extension services, language barriers, and weak information channels compound the problem, leaving urgent agronomic issues to linger alongside routine ones.

This research proposes a retrieval system built for a Retrieval-Augmented Generation (RAG) pipeline. Given a farmer's natural language query describing symptoms or field conditions, the system returns the five most relevant documents from a curated agricultural knowledge base. The approach is guided by relevance to real farmer language rather than exact keyword matching, by localization to Nigeria's major crops such as cassava, maize, and rice, and by rigorous benchmarking against expert-labeled judgments rather than assumptions about what counts as a good result.

Success means a farmer can describe a problem in plain language and reliably find the document that answers it, extension workers spend less time searching through irrelevant material, and the most relevant guidance consistently surfaces at the top of the results.

Word count: approximately 220 words.